Chapter 1

Python Classes

FIXME integrate this better with polynomials, but we need to cover the following

1.1 Object-Oriented Programming Introduction

Imagine you want to implement multiple dogs in python. It gets a bit complicated to do the following:

```
    dog1name = "Teddy"
    dog1age = 6
    dog1sound = "woof"

    dog2name = "Fluffers"
    dog2age = 2
    dog2sound = "bark"

    dog3name = "Bella"
    dog3age = 3
    dog3sound = "grr"

    print(f"{dog1name} is {dog1age} and says {dog1sound}!")
    print(f"{dog2name} is {dog2age} and says {dog2sound}!")
    print(f"{dog3name} is {dog3age} and says {dog3sound}!")


```

Instead, we can use classes, which are a way to create your own datatype. Classes can contain custom methods that either return, print, or calculate different values for you, and they can contain custom variables referred to as attributes.

```
class Dog:
    """A simple model of a dog."""
    # constructor - which creates a new model of a dog.
    def __init__(self, name: str, age: int, sound: str):
        self.name = name
        self.age = age
        self.sound = sound
    # method speak
    def speak(self) -> None:
```

```
        """Print a sentence describing the dog."""
        print(f"{self.name} is {self.age} and says {self.sound}!")

# create dog objects called "instances" with the given variables
dog1 = Dog("Teddy", 6, "woof")
dog2 = Dog("Fluffers", 2, "bark")
dog3 = Dog("Bella", 3, "grr")

# call the behavior on each object
dog1.speak()
dog2.speak()
dog3.speak()

```

Now let's talk about classes in the context of polynomials!

The built-in types, such as strings, have functions associated with them. So, for example, if you needed a string converted to uppercase, you would call its `upper()` function: -

```
my_string = "houston, we have a problem!"
louder_string = my_string.upper()
```

This would set `louder_string` to "HOUSTON, WE HAVE A PROBLEM!" When a function is associated with a datatype like this, it is called a *method*. A datatype with methods is known as a *class*. Creating a new version of a class in a variable is called an *instance*. For example, in the example, we would say "`my_string` is an instance of the class `str`. `str` has a method called `upper`"

The function `type` will tell you the type of any data:

```
print(type(my_string))
```

This will output

```
<class 'str'>
```

A class can also define operators. `+`, for example, is redefined by `str` to concatenate strings together:

```
long_string = "I saw " + "15 people"
```

1.2 Parent classes

Classes can have classes they inherit from (called "parent classes") or classes that inherit from them (called "child classes"). Parent classes (ie 'Animal') give a subclass (or child) different attributes or methods. Let's check out an example:

```
class Animal:
    """Generic animal base-class."""

    def __init__(self, name: str, age: int) -> None:
        self.name = name
        self.age = age

    def describe(self) -> None:
        """Print a basic description common to all animals."""
        print(f"{self.name} is {self.age} years old.")

class Dog(Animal):
    """Dog inherits name and age from Animal, adds its own sound."""

    def __init__(self, name: str, age: int, sound: str) -> None:
        super().__init__(name, age)      # initialize the Animal part
        self.sound = sound

    def speak(self) -> None:
        """Dog-specific implementation of speak()."""
        print(f"{self.name} is {self.age} and says {self.sound}!")


# demonstration
dogs = [
    Dog("Teddy", 6, "woof"),
    Dog("Fluffers", 2, "bark"),
    Dog("Bella", 3, "grr")
]

for dog in dogs:
    dog.describe()    # common behavior from Animal
    dog.speak()       # overridden behavior in Dog

```

Here, we have made a parent class for 'dog' called 'animal'.

1.3 Making a Polynomial class

You have created a bunch of useful python functions for dealing with polynomials. Notice how each one has the word "polynomial" in the function name like `derivative_of_polynomial`.

Wouldn't it be more elegant if you had a Polynomial class with a `derivative` method? Then you could use your polynomial like this:

```
a = Polynomial([9.0, 0.0, 2.3])
b = Polynomial([-2.0, 4.5, 0.0, 2.1])

print(a, "plus", b , "is", a+b)
print(a, "times", b , "is", a*b)
print(a, "times", 3 , "is", a*3)
print(a, "minus", b , "is", a-b)

c = b.derivative()

print("Derivative of", b ,"is", c)
```

And it would output:

```
2.30x^2 + 9.00 plus 2.10x^3 + 4.50x + -2.00 is 2.10x^3 + 2.30x^2 + 4.50x + 7.00
2.30x^2 + 9.00 times 2.10x^3 + 4.50x + -2.00 is 4.83x^5 + 29.25x^3 + -4.60x^2 +
↪  40.50x + -18.00
2.30x^2 + 9.00 times 3 is 6.90x^2 + 27.00
2.30x^2 + 9.00 minus 2.10x^3 + 4.50x + -2.00 is -2.10x^3 + 2.30x^2 + -4.50x +
↪  11.00
Derivative of 2.10x^3 + 4.50x + -2.00 is 6.30x^2 + 4.50
```

Create a file for your class definition called `Polynomial.py`. Enter the following:

```
class Polynomial:
    def __init__(self, coeffs):
        self.coefficients = coeffs.copy()

    def __repr__(self):
        # Make a list of the monomial strings
        monomial_strings = []

        # For standard form we start at the largest degree
        degree = len(self.coefficients) - 1

        # Go through the list backwards
        while degree >= 0:
            coefficient = self.coefficients[degree]

            if coefficient != 0.0:
                # Describe the monomial
                if degree == 0:
                    monomial_string = "{:.2f}".format(coefficient)
                elif degree == 1:
                    monomial_string = "{:.2f}x".format(coefficient)
```

```
22                  else:
23                      monomial_string = "{:.2f}x^{}".format(coefficient, degree)
24
25                  # Add it to the list
26                  monomial_strings.append(monomial_string)
27
28              # Move to the previous term
29              degree = degree - 1
30
31          # Deal with the zero polynomial
32          if len(monomial_strings) == 0:
33              monomial_strings.append("0.0")
34
35          # Separate the terms with a plus sign
36          return " + ".join(monomial_strings)
37
38      def __call__(self, x):
39          sum = 0.0
40          for degree, coefficient in enumerate(self.coefficients):
41              sum = sum + coefficient * x ** degree
42          return sum
43
44      def __add__(self, b):
45          result_length = max(len(self.coefficients), len(b.coefficients))
46          result = []
47          for i in range(result_length):
48              if i < len(self.coefficients):
49                  coefficient_a = self.coefficients[i]
50              else:
51                  coefficient_a = 0.0
52
53              if i < len(b.coefficients):
54                  coefficient_b = b.coefficients[i]
55              else:
56                  coefficient_b = 0.0
57              result.append(coefficient_a + coefficient_b)
58
59          return Polynomial(result)
60
61      def __mul__(self, other):
62
63          # Not a polynomial?
64          if not isinstance(other, Polynomial):
65              # Try to make it a constant polynomial
66              other = Polynomial([other])
67
68          # What is the degree of the resulting polynomial?
69          result_degree = (len(self.coefficients) - 1) + (len(other.coefficients) -
            ↪ 1)
70
71          # Make a list of zeros to hold the coefficients
72          result = [0.0] * (result_degree + 1)
```

```
        # Iterate over the indices and values of a
        for a_degree, a_coefficient in enumerate(self.coefficients):

            # Iterate over the indices and values of b
            for b_degree, b_coefficient in enumerate(other.coefficients):

                # Calculate the resulting monomial
                coefficient = a_coefficient * b_coefficient
                degree = a_degree + b_degree

                # Add it to the right bucket
                result[degree] = result[degree] + coefficient

        return Polynomial(result)

    __rmul__ = __mul__

    def __sub__(self, other):
        return self + other * -1.0

    def derivative(self):

        # What is the degree of the resulting polynomial?
        original_degree = len(self.coefficients) - 1
        if original_degree > 0:
            degree_of_derivative = original_degree - 1
        else:
            degree_of_derivative = 0

        # We can ignore the constant term (skip the first coefficient)
        current_degree = 1
        result = []

        # Differentiate each monomial
        while current_degree < len(self.coefficients):
            coefficient = self.coefficients[current_degree]
            result.append(coefficient * current_degree)
            current_degree = current_degree + 1

        # No terms? Make it the zero polynomial
        if len(result) == 0:
            result.append(0.0)

        return Polynomial(result)
```

Create a second file called `test_polynomial.py` to test it:

```
from Polynomial import Polynomial

```

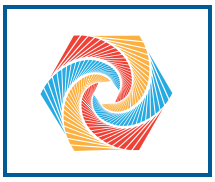
```
a = Polynomial([9.0, 0.0, 2.3])
b = Polynomial([-2.0, 4.5, 0.0, 2.1])

print(a, "plus", b , "is", a+b)
print(a, "times", b , "is", a*b)
print(a, "times", 3 , "is", a*3)
print(a, "minus", b , "is", a-b)

c = b.derivative()

print("Derivative of", b ,"is", c)

slope = c(3)
print("Value of the derivative at 3 is", slope)

```

Run the test code:

```
python3 test_polynomial.py
```

Appendix A

Answers to Exercises



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